Change of Axes

Transformation of Co-ordinates:
The co-ordinates of a point or the equation of a curve are always given with reference to a fixed origin and a set of axes of co-ordinates. The above co-ordinates of the equation of the curve changes when the origin is changed or the direction of axes changed or both. The process of changing the co-ordinate of a point or the equation of a curve is called transformation of co-ordinates.

1. **Determine the equation of the curve $2x^2 + 3y^2 - 8x + 6y - 7 = 0$ when the origin is transferred to the point $(2, -1)$.**

Solution:
Given equation is $2x^2 + 3y^2 - 8x + 6y - 7 = 0$…….$(i)$

When the origin is transferred to the point $(2, -1)$

Put $x = 2 + x'$ and $y = -1 + y'$

Putting the values of x and y in (i), we get

$2(2 + x')^2 + 3(-1 + y')^2 - 8(2 + x') + 6(-1 + y') - 7 = 0$

$\Rightarrow 2(x'^2 + 4x' + 4) + 3(y'^2 - 2y' + 1) - 8(2 + x') + 6(-1 + y') - 7 = 0$

$\Rightarrow 2x'^2 + 8x' + 8 + 3y'^2 - 6y' + 3 - 16 - 8x' - 6 + 6y' - 7 = 0$

$\Rightarrow 2x'^2 + 3y'^2 - 18 = 0$

$\Rightarrow 2x'^2 + 3y'^2 = 18$

Now removing suffixes, the equation referred to new axes is

$2x^2 + 3y^2 = 18$

2. Determine the equation of the curve $x^2 + 2y^2 - 6x + 7 = 0$ when the origin is transferred to the point $(3, 1)$.

Solution:

Given equation is $x^2 + 2y^2 - 6x + 7 = 0$ …….(i)

When the origin is transferred to the point $(3, 1)$

Put $x = 3 + x'$ and $y = 1 + y'$

Putting the values of x and y in (i), we get

$(3 + x')^2 + 2(1 + y')^2 - 6(3 + x') + 7 = 0$

$\Rightarrow (x'^2 + 6x' + 9) + 2(y'^2 + 2y' + 1) - 6(3 + x') + 7 = 0$

	$\Rightarrow x'^2 + 6x' + 9 + 2y'^2 + 4y' + 2 - 18 - 6x' + 7 = 0$ $\Rightarrow x'^2 + 2y'^2 + 4y' = 0$ Now removing suffixes, the equation referred to new axes is $x^2 + 2y^2 + 4y = 0$
3.	**Determine the equation of the parabola $x^2 - 2xy + y^2 + 2x - 4y + 3 = 0$ after rotating of axes throw 45^0.**
	Solution: Given equation is $x^2 - 2xy + y^2 + 2x - 4y + 3 = 0$ ……(i) If we rotate the axes through an angle 45^0, then we get $x = x'cos45^0 - y'sin45^0 = \frac{x' - y'}{\sqrt{2}}$ $y = x'sin45^0 + y'cos45^0 = \frac{x' + y'}{\sqrt{2}}$ Putting the values of x and y in (i), we get $\left(\frac{x' - y'}{\sqrt{2}}\right)^2 - 2\left(\frac{x' - y'}{\sqrt{2}}\right)\left(\frac{x' + y'}{\sqrt{2}}\right) + \left(\frac{x' + y'}{\sqrt{2}}\right)^2 + 2\left(\frac{x' - y'}{\sqrt{2}}\right) - 4\left(\frac{x' + y'}{\sqrt{2}}\right) + 3 = 0$ $\Rightarrow \frac{(x' - y')^2}{2} - (x' - y')(x' + y') + \frac{(x' + y')^2}{2} + \sqrt{2}(x' - y') - 2\sqrt{2}(x' + y') + 3 = 0$ $\Rightarrow (x' - y')^2 - 2(x'^2 - y'^2) + (x' + y')^2 + 2\sqrt{2}(x' - y') - 4\sqrt{2}(x' + y') + 6 = 0$ $\Rightarrow x'^2 - 2x'y' + y'^2 - 2x'^2 + 2y'^2 + x'^2 + 2x'y' + y'^2 + 2\sqrt{2}x' - 2\sqrt{2}y' - 4\sqrt{2}x' - 4\sqrt{2}y' + 6 = 0$ $\Rightarrow 4y'^2 - 2\sqrt{2}x' - 6\sqrt{2}y' + 6 = 0$ $\Rightarrow 2y'^2 - \sqrt{2}x' - 3\sqrt{2}y' + 3 = 0$ Now removing suffixes, the equation referred to new axes is $\Rightarrow 2y^2 - \sqrt{2}x - 3\sqrt{2}y + 3 = 0$

4.	**Determine the equation of the parabola $x^2 - y^2 - 2\sqrt{2}\,x - 10\sqrt{2}\,y + 2 = 0$ after rotating of axes throw 45^0.**
	Solution: Given equation is $x^2 - y^2 - 2\sqrt{2}\,x - 10\sqrt{2}\,y + 2 = 0$ ……..(i) If we rotate the axes through an angle 45^0, then we get $x = x'cos45^0 - y'sin45^0 = \frac{x' - y'}{\sqrt{2}}$ $y = x'sin45^0 + y'cos45^0 = \frac{x' + y'}{\sqrt{2}}$ Putting the values of x and y in (i), we get $\left(\frac{x' - y'}{\sqrt{2}}\right)^2 - \left(\frac{x' + y'}{\sqrt{2}}\right)^2 - 2\sqrt{2}\left(\frac{x' - y'}{\sqrt{2}}\right) - 10\sqrt{2}\left(\frac{x' + y'}{\sqrt{2}}\right) + 2 = 0$ $\Rightarrow \frac{(x' - y')^2}{2} - \frac{(x' + y')^2}{2} - 2(x' - y') - 10(x' + y') + 2 = 0$ $\Rightarrow (x' - y')^2 - (x' + y')^2 - 4(x' - y') - 20(x' + y') + 4 = 0$ $\Rightarrow x'^2 - 2x'y' + y'^2 - x'^2 - 2x'y' - y'^2 - 4x' + 4y' - 20x' - 20y' + 4 = 0$ $\Rightarrow -4x'y' - 24x' - 16y' + 4 = 0$ Now removing suffixes, the equation referred to new axes is $\Rightarrow -4xy - 24x - 16y + 4 = 0$
5.	**If the direction of axes is turned through an angle 30^0 and the origin remains unchanged then find the transformation equation of $x^2 + 2\sqrt{3}xy - y^2 - 2a^2 = 0$.**
	Solution: Given equation is $x^2 + 2\sqrt{3}xy - y^2 - 2a^2 = 0$. ……..$(i)$ If we rotate the axes through an angle 30^0, then we get

	$x = x'cos30^0 - y'sin30^0 = \frac{\sqrt{3}}{2}x' - \frac{1}{2}y' = \frac{\sqrt{3}x' - y'}{2}$ $y = x'sin30^0 + y'cos30^0 = \frac{1}{2}x' - \frac{\sqrt{3}}{2}y' = \frac{x' + \sqrt{3}y'}{2}$ Putting the values of x and y in (i), we get $\left(\frac{\sqrt{3}x' - y'}{2}\right)^2 + 2\sqrt{3}\left(\frac{\sqrt{3}x' - y'}{2}\right)\left(\frac{x' + \sqrt{3}y'}{2}\right) - \left(\frac{x' + \sqrt{3}y'}{2}\right)^2 - 2a^2 = 0$ $\Rightarrow \frac{\left(\sqrt{3}x' - y'\right)^2}{4} + \frac{2\sqrt{3}}{4}\left(\sqrt{3}x'^2 + 3x'y' - x'y' - \sqrt{3}y'^2\right) - \frac{\left(x' + \sqrt{3}y'\right)^2}{4} - 2a^2 = 0$ $\Rightarrow \left(\sqrt{3}x' - y'\right)^2 + 2\sqrt{3}\left(\sqrt{3}x'^2 + 3x'y' - x'y' - \sqrt{3}y'^2\right) - \left(x' + \sqrt{3}y'\right)^2 - 8a^2 = 0$ $\Rightarrow 3x'^2 - 2\sqrt{3}x'y' + y'^2 + 6x'^2 + 6\sqrt{3}x'y' - 2\sqrt{3}x'y' - 6y'^2 - x'^2 - 2\sqrt{3}x'y' - 3y'^2 - 8a^2 = 0$ $\Rightarrow 8x'^2 - 8y'^2 - 8a^2 = 0$ $\Rightarrow x'^2 - y'^2 - a^2 = 0$ Now removing suffixes, the equation referred to new axes is $\Rightarrow x^2 - y^2 - a^2 = 0$
6.	**Transform the equation $11x^2 + 24xy + 4y^2 - 20x - 40y - 5 = 0$ to rectangular axes through the point $(2, -1)$ and inclined at an angle $tan^{-1}\left(-\frac{4}{3}\right)$.**
	Solution: Given equation is $11x^2 + 24xy + 4y^2 - 20x - 40y - 5 = 0$ ……(i) When the origin is transferred to the point $(2, -1)$ Put $x = 2 + x'$ and $y = -1 + y'$ Putting the values of x and y in (i), we get $11(2 + x')^2 + 24(2 + x')(-1 + y') + 4(-1 + y')^2 - 20(2 + x') - 40(-1 + y') - 5 = 0$ $\Rightarrow 11(x'^2 + 4x' + 4) + 24(-2 + 2y' - x' + x'y') + 4(y'^2 - 2y' + 1) - 20(2 + x') - 40(-1 + y') - 5 = 0$

$\Rightarrow 11x'^2 + 44x' + 44 - 48 + 48y' - 24x' + 24x'y' + 4y'^2 - 8y' + 4 - 40 - 20x' + 40 - 40y' - 5 = 0$

$\Rightarrow 11x'^2 + 24x'y' + 4y'^2 - 5 = 0$

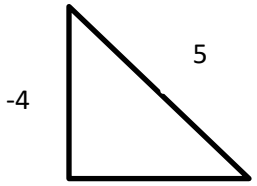
Now removing suffixes, the equation referred to new axes is

$11x^2 + 24xy + 4y^2 - 5 = 0 \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots (ii)$

For the second transform $tan\theta = -\frac{4}{3}$

Also $sin\theta = -\frac{4}{5}$ and $cos\theta = \frac{3}{5}$

Now put



$x = x'cos\theta - y'sin\theta = \frac{3x'}{5} + \frac{4y'}{5} = \frac{3x' + 4y'}{5}$

$y = x'sin\theta + y'cos\theta = \frac{3y'}{5} - \frac{4x'}{5} = \frac{3y' - 4x'}{5}$

Putting the values of x and y in (ii), we get

$$11\left(\frac{3x' + 4y'}{5}\right)^2 + 24\left(\frac{3x' + 4y'}{5}\right)\left(\frac{3y' - 4x'}{5}\right) + 4\left(\frac{3y' - 4x'}{5}\right)^2 - 5 = 0$$

$$\Rightarrow 11\frac{(3x' + 4y')^2}{25} + 24\frac{\left(-12x'^2 + 12y'^2 + 9x'y' - 16x'y'\right)}{25} + 4\frac{(3y' - 4x')^2}{25} - 5 = 0$$

$$\Rightarrow 11(3x' + 4y')^2 + 24\left(-12x'^2 + 12y'^2 - 7x'y'\right) + 4(3y' - 4x')^2 - 125 = 0$$

$$\Rightarrow 11\left(9x'^2 + 16y'^2 + 24x'y'\right) + 24\left(-12x'^2 + 12y'^2 - 7x'y'\right) + 4\left(9y'^2 + 16x'^2 - 24x'y'\right) - 125 = 0$$

$$\Rightarrow 99x'^2 + 176y'^2 + 264x'y' - 288x'^2 + 288y'^2 - 168x'y' + 36y'^2 + 64x'^2 - 96x'y' - 125 = 0$$

$$\Rightarrow -125x'^2 + 500y'^2 - 125 = 0$$

$$\Rightarrow x'^2 - 4y'^2 + 1 = 0$$

Now removing suffixes, the equation referred to new axes is

$$\Rightarrow x^2 - 4y^2 + 1 = 0$$

7.	**Transform the equation $x^2 - 4xy + y^2 + 8x - 2y - 5 = 0$ to new axes through $(2, 3)$ rotated through an angle of 45^0.**
	Solution: Given equation is $x^2 - 4xy + y^2 + 8x - 2y - 5 = 0$ ……(i) When the origin is transferred to the point $(2, 3)$ and rotate the axes through an angle 45^0, then we get $x = 2 + x'cos45^0 - y'sin45^0 = 2 + \frac{x' - y'}{\sqrt{2}}$ $y = 3 + x'sin45^0 + y'cos45^0 = 3 + \frac{x' + y'}{\sqrt{2}}$ Putting the values of x and y in (i), we get $\left(2 + \frac{x' - y'}{\sqrt{2}}\right)^2 - 4\left(2 + \frac{x' - y'}{\sqrt{2}}\right)\left(3 + \frac{x' + y'}{\sqrt{2}}\right) + \left(3 + \frac{x' + y'}{\sqrt{2}}\right)^2 + 8\left(2 + \frac{x' - y'}{\sqrt{2}}\right) - 2\left(3 + \frac{x' + y'}{\sqrt{2}}\right) - 5 = 0$ $\Rightarrow 4 + 4.\frac{x' - y'}{\sqrt{2}} + \frac{(x' - y')^2}{2} - 4\left(\frac{2\sqrt{2} + x' - y'}{\sqrt{2}}\right)\left(\frac{3\sqrt{2} + x' + y'}{\sqrt{2}}\right) + 9 + 6.\frac{x' + y'}{\sqrt{2}} + \frac{(x' + y')^2}{2} + 8\left(\frac{2\sqrt{2} + x' - y'}{\sqrt{2}}\right) - 2\left(\frac{3\sqrt{2} + x' + y'}{\sqrt{2}}\right) - 5 = 0$ $\Rightarrow 8 + 4\sqrt{2}(x' - y') + (x' - y')^2 - 4\left(2\sqrt{2} + x' - y'\right)\left(3\sqrt{2} + x' + y'\right) + 18 + 6\sqrt{2}(x' + y') + (x' + y')^2 + 8\sqrt{2}\left(2\sqrt{2} + x' - y'\right) - 2\sqrt{2}\left(3\sqrt{2} + x' + y'\right) - 10 = 0$ $\Rightarrow 8 + 4\sqrt{2}x' - 4\sqrt{2}y' + x'^2 - 2x'y' + y'^2 - 4\left(12 + 2\sqrt{2}x' + 2\sqrt{2}y' + 3\sqrt{2}x' + x'^2 + x'y' - 3\sqrt{2}y' - x'y' - y'^2\right) + 18 + 6\sqrt{2}x' + 6\sqrt{2}y' + x'^2 + 2x'y' + y'^2 + 32 + 8\sqrt{2}x' - 8\sqrt{2}y' - 12 - 2\sqrt{2}x' - 2\sqrt{2}y' - 10 = 0$

	$\Rightarrow 8+4\sqrt{2}x'-4\sqrt{2}y'+x'^2-2x'y'+y'^2-48-8\sqrt{2}x'-8\sqrt{2}y'-12\sqrt{2}x'$ $-4x'^2-4x'y'+12\sqrt{2}y'+4x'y'+4y'^2+18+6\sqrt{2}x'$ $+6\sqrt{2}y'+x'^2+2x'y'+y'^2+32+8\sqrt{2}x'-8\sqrt{2}y'-12-2\sqrt{2}x'$ $-2\sqrt{2}y'-5=0$ $\Rightarrow -2x'^2+6y'^2+\sqrt{2}x'(4-8-12+6+8-2)-\sqrt{2}y'(4+8-12-6+8+2)+8-48$ $+18+32-12-10=0$ $\Rightarrow -2x'^2+6y'^2-4\sqrt{2}x'-4\sqrt{2}y'-12=0$ $\Rightarrow x'^2-3y'^2+2\sqrt{2}x'+2\sqrt{2}y'+6=0$ Now removing suffixes, the equation referred to new axes is $\Rightarrow x^2-3y^2+2\sqrt{2}x+2\sqrt{2}y+6=0$
8.	**Invariants:** **If by the rotation of the rectangular co-ordinate axes about the origin, the expression $ax^2+2hxy+by^2$ change to $a'x'^2+2h'x'y'+b'y'^2$ then $a+b=a'+b'$ and $ab-h^2=a'b'-h'^2$.**
	Solution: If the axes be rotated though an angle θ about the origin Then $x=x'cos\theta-y'sin\theta$ $y=x'sin\theta+y'cos\theta$ Substituting these values, the expression $ax^2+2hxy+by^2$ becomes $a(x'cos\theta-y'sin\theta)^2+2h(x'cos\theta-y'sin\theta)(x'sin\theta+y'cos\theta)+b(x'sin\theta+y'cos\theta)^2$ $=ax'^2cos^2\theta-2ax'y'sin\theta\,cos\theta+ay'^2sin^2\theta+2hx'^2sin\theta\,cos\theta+2hx'y'\,cos^2\theta$ $-2hx'y'sin^2\theta-2h\,y'^2sin\theta\,cos\theta+bx'^2sin^2\theta+2bx'y'sin\theta\,cos\theta$ $+by'^2cos^2\theta$ $=x'^2(a\,cos^2\theta+2h\,sin\theta\,cos\theta+bsin^2\theta)$ $+2x'y'(-a\,sin\theta\,cos\theta+bsin\theta\,cos\theta+h\,cos^2\theta-h\,sin^2\theta)$ $+y'^2(a\,sin^2\theta-2h\,sin\theta\,cos\theta+bcos^2\theta)$ $=a'x'^2+2h'x'y'+b'y'^2$ Where $a'=a\,cos^2\theta+2h\,sin\theta\,cos\theta+bsin^2\theta$

$b' = a\,sin^2\theta - 2h\,sin\theta\,cos\theta + bcos^2\theta$

$h' = -a\,sin\theta\,cos\theta + bsin\theta\,cos\theta + h\,cos^2\theta - h\,sin^2\theta$

$\therefore a' + b' = a\,cos^2\theta + 2h\,sin\theta\,cos\theta + bsin^2\theta + a\,sin^2\theta - 2h\,sin\theta\,cos\theta + bcos^2\theta$

$= a(sin^2\theta + cos^2\theta) + b(sin^2\theta + cos^2\theta)$

$\Rightarrow a' + b' = a + b$

Now

$2a' = 2a\,cos^2\theta + 2.2h\,sin\theta\,cos\theta + 2b\,sin^2\theta$

$= a(1 + cos2\theta) + 2h\,sin2\theta\ + b(1 - cos2\theta)$

$= (a + b) + \{(a - b)cos2\theta + 2h\,sin2\theta\}$

$2b' = 2a\,sin^2\theta - 2.2h\,sin\theta\,cos\theta + 2b\,cos^2\theta$

$= a(1 - cos2\theta) - 2h\,sin2\theta\ + b(1 + cos2\theta)$

$= (a + b) - \{(a - b)cos2\theta + 2h\,sin2\theta\}$

$h' = -a\,sin\theta\,cos\theta + bsin\theta\,cos\theta + h\,cos^2\theta - h\,sin^2\theta$

$= -(a - b)sin\theta\,cos\theta + h(\,cos^2\theta - \,sin^2\theta)$

$2h' = -\{(a - b)2sin\theta\,cos\theta - 2h\,(\,cos^2\theta - \,sin^2\theta)\}$

$2h' = -\{(a - b)sin2\theta - 2h\,cos2\theta\}$

Now

$4a'b' - 4h'^2 = (a + b)^2 - \{(a - b)cos2\theta + 2h\,sin2\theta\}^2 - \{(a - b)sin2\theta\ - 2h\,cos2\theta\}^2$

$= (a + b)^2 - (a - b)^2cos^2 2\theta - 4h\,sin2\theta\,cos2\theta\,(a - b) - 4h^2sin^2 2\theta - (a - b)^2sin^2 2\theta + 4h\,sin2\theta\,cos2\theta\,(a - b) - 4h^2cos^2 2\theta$

$= -4h^2(sin^2 2\theta + cos^2 2\theta) - (a - b)^2(sin^2 2\theta + cos^2 2\theta) + (a + b)^2$

$4a'b' - 4h'^2 = -4h^2 + 4ab$

$\therefore a'b' - h'^2 = ab - h^2$

9.	**Remove the xy term from the expression $ax^2 + 2hxy + by^2$** **OR** **Show that if a set of rectangular axes be turned through an angle θ about the origin, then the expression $ax^2 + 2hxy + by^2$ transformed to the form $a'x'^2 + b'y'^2$ in a new co-ordinate if $\theta = \frac{1}{2}tan^{-1}\left(\frac{2h}{a-b}\right)$.**
	Solution: If the axes be rotated though an angle θ about the origin Then $x = x'cos\theta - y'sin\theta$ $y = x'sin\theta + y'cos\theta$ Substituting these values, the expression $ax^2 + 2hxy + by^2$ becomes $a(x'cos\theta - y'sin\theta)^2 + 2h(x'cos\theta - y'sin\theta)(x'sin\theta + y'cos\theta) + b(x'sin\theta + y'cos\theta)^2$ $= ax'^2cos^2\theta - 2ax'y'sin\theta\ cos\theta + ay'^2sin^2\theta + 2hx'^2sin\theta\ cos\theta + 2hx'y'\ cos^2\theta$ $- 2hx'y'sin^2\theta - 2h\ y'^2sin\theta\ cos\theta + bx'^2sin^2\theta + 2bx'y'sin\theta\ cos\theta$ $+ by'^2cos^2\theta$ $= x'^2(a\ cos^2\theta + 2h\ sin\theta\ cos\theta + bsin^2\theta)$ $+ 2x'y'(-a\ sin\theta\ cos\theta + bsin\theta\ cos\theta + h\ cos^2\theta - h\ sin^2\theta)$ $+ y'^2(a\ sin^2\theta - 2h\ sin\theta\ cos\theta + bcos^2\theta)$ $= a'x'^2 + 2h'x'y' + b'y'^2$ Where $a' = a\ cos^2\theta + 2h\ sin\theta\ cos\theta + bsin^2\theta$ $b' = a\ sin^2\theta - 2h\ sin\theta\ cos\theta + bcos^2\theta$ $h' = -a\ sin\theta\ cos\theta + bsin\theta\ cos\theta + h\ cos^2\theta - h\ sin^2\theta$ The xy term will be removed from the given expression if $h' = 0$ $i,e;\ -\{(a-b)sin\theta\ cos\theta - h(\ cos^2\theta - \ sin^2\theta)\} = 0$ $\Rightarrow (a-b)sin\theta\ cos\theta - h(\ cos^2\theta - \ sin^2\theta) = 0$ $\Rightarrow (a-b)sin\theta\ cos\theta = h(\ cos^2\theta - \ sin^2\theta)$ $\Rightarrow 2(a-b)sin\theta\ cos\theta = 2h(\ cos^2\theta - \ sin^2\theta)$ $\Rightarrow (a-b)sin2\theta\ = 2hcos2\theta$

	$\Rightarrow \frac{sin2\theta}{cos2\theta} = \frac{2h}{(a-b)}$ $\Rightarrow tan2\theta = \frac{2h}{(a-b)}$ $\Rightarrow 2\theta = tan^{-1}\frac{2h}{(a-b)}$ $\therefore \theta = \frac{1}{2}\tan^{-1}\left(\frac{2h}{a-b}\right)$ Hence if the axes are rotated about the origin through an angle $\frac{1}{2}\tan^{-1}\left(\frac{2h}{a-b}\right)$, then the xy term from the expression $ax^2 + 2hxy + by^2$ will be removed.
10	**Determine the angle through which the axes must be rotated to remove the xy term in the equation $7x^2 - 6\sqrt{3}xy + 13y^2 = 16$.**
	Solution: Given that $7x^2 - 6\sqrt{3}xy + 13y^2 = 16$ (i) If the axes be rotated though an angle θ about the origin Then $x = x'cos\theta - y'sin\theta$ $y = x'sin\theta + y'cos\theta$ Substituting these values, the expression $7x^2 - 6\sqrt{3}xy + 13y^2 = 16$ becomes $7(x'cos\theta - y'sin\theta)^2 - 6\sqrt{3}(x'cos\theta - y'sin\theta)(x'sin\theta + y'cos\theta) + 13(x'sin\theta + y'cos\theta)^2 = 16$ $\Rightarrow x'^2\left(7\,cos^2\theta - 6\sqrt{3}\,sin\theta\,cos\theta + 13\,sin^2\theta\right) + 2x'y'\left(-7\,sin\theta\,cos\theta + 13\,sin\theta\,cos\theta - 3\sqrt{3}\,cos^2\theta + 3\sqrt{3}\,sin^2\theta\right) + y'^2\left(7\,sin^2\theta + 6\sqrt{3}\,sin\theta\,cos\theta + 13\,cos^2\theta\right) = 16$ $\Rightarrow a'x'^2 + 2h'x'y' + b'y'^2 = 16$ (ii) Where $a' = 7\,cos^2\theta - 6\sqrt{3}\,sin\theta\,cos\theta + 13\,sin^2\theta$ $b' = 7\,sin^2\theta + 6\sqrt{3}\,sin\theta\,cos\theta + 13\,cos^2\theta$ $h' = -7\,sin\theta\,cos\theta + 13\,sin\theta\,cos\theta - 3\sqrt{3}\,cos^2\theta + 3\sqrt{3}\,sin^2\theta$ Making the co-efficient of xy term is zero i.e., $h' = 0$

$-7\ sin\theta\ cos\theta + 13\ sin\theta\ cos\theta - 3\sqrt{3}\ cos^2\theta + 3\sqrt{3}\ sin^2\theta = 0$

$\Rightarrow 6\ sin\theta\ cos\theta - 3\sqrt{3}(\ cos^2\theta - \ sin^2\theta) = 0$

$\Rightarrow 3\ sin2\theta - 3\sqrt{3}\ cos2\theta = 0$

$\Rightarrow 3\ sin2\theta = 3\sqrt{3}\ cos2\theta$

$\Rightarrow \ \frac{sin2\theta}{cos2\theta} = \sqrt{3}$

$\Rightarrow \ tan2\theta = tan60^0$

$\Rightarrow \ 2\theta = 60^0$

$\therefore \theta = 30^0$

Now the equation (ii) becomes

$\therefore a'x'^2 + b'y'^2 = 16$ (iii)

Now

$\Rightarrow a' = 7\ cos^2\theta - 6\sqrt{3}\ sin\theta\ cos\theta + 13\ sin^2\theta$

$\Rightarrow a' = 7(cos30^0)^2 - 6\sqrt{3}\ sin30^0\ cos30^0 + 13\ (sin30^0)^2$

$\Rightarrow a' = 7\left(\frac{\sqrt{3}}{2}\right)^2 - 6\sqrt{3}\ .\frac{1}{2}\ .\frac{\sqrt{3}}{2} + 13\ \left(\frac{1}{2}\right)^2$

$\Rightarrow a' = 7\frac{3}{4} - 6\sqrt{3}\ .\frac{1}{2}\ .\frac{\sqrt{3}}{2} + 13\ \frac{1}{4}$

$\Rightarrow a' = \frac{21}{4} - \frac{18}{4} + \frac{13}{4}$

$\Rightarrow a' = \frac{21 - 18 + 13}{4}$

$\Rightarrow a' = \frac{16}{4}$

$\Rightarrow a' = 4$

Again

$\Rightarrow b' = 7\ sin^2\theta + 6\sqrt{3}\ sin\theta\ cos\theta + 13\ cos^2\theta$

	$\Rightarrow b' = 7(sin30^0)^2 + 6\sqrt{3}\, sin30^0\, cos30^0 + 13\,(cos30^0)^2$ $\Rightarrow b' = 7\left(\frac{1}{2}\right)^2 + 6\sqrt{3}\,.\frac{1}{2}\,.\frac{\sqrt{3}}{2} + 13\,\left(\frac{\sqrt{3}}{2}\right)^2$ $\Rightarrow b' = 7\frac{1}{4} + 6\sqrt{3}\,.\frac{1}{2}\,.\frac{\sqrt{3}}{2} + 13\,\frac{3}{4}$ $\Rightarrow b' = \frac{7}{4} + \frac{18}{4} + \frac{39}{4}$ $\Rightarrow b' = \frac{7 + 18 + 39}{4}$ $\Rightarrow b' = \frac{64}{4}$ $\Rightarrow b' = 16$ Putting the values a' and b' in (iii), we get $4x'^2 + 16y'^2 = 16$ Now removing suffixes, the equation is $4x^2 + 16y^2 = 16$ $\therefore x^2 + 4y^2 = 4$
11.	**Remove the first-degree terms in $3x^2 - 4y^2 - 6x - 8y - 10 = 0$**
	Solution: Given equation is $3x^2 - 4y^2 - 6x - 8y - 10 = 0$……$(i)$ To remove x and y terms shift the origin at (α, β) without change of direction of the axes. Put $x = \alpha + x'$ and $y = \beta + y'$ Putting the values of x and y in (i), we get $3(\alpha + x')^2 - 4(\beta + y')^2 - 6(\alpha + x') - 8(\beta + y') - 10 = 0$ $\Rightarrow 3(x'^2 + 2\alpha x' + \alpha^2) - 4(y'^2 + 2\beta y' + \beta^2) - 6(\alpha + x') - 8(\beta + y') - 10 = 0$ $\Rightarrow 3x'^2 + 6\alpha x' + 3\alpha^2 - 4y'^2 - 8\beta y' - 4\beta^2 - 6\alpha - 6x' - 8\beta - 8y' - 10 = 0$ $\Rightarrow 3x'^2 - 4y'^2 + (6\alpha - 6)x' - (8\beta + 8)y' + 3\alpha^2 - 4\beta^2 - 6\alpha - 8\beta - 10 = 0 \ldots\ldots (ii)$

	To remove x and y terms Put $6\alpha - 6 = 0$ and $8\beta + 8 = 0$ $\therefore \alpha = 1$ and $\beta = -1$ Putting the values of α and β in (ii), we get $\Rightarrow 3x'^2 - 4y'^2 + (6.1 - 6)x' - \{8.(-1) + 8\}y' + 3(1)^2 - 4(-1)^2 - 6.1 - 8(-1) - 10 = 0$ $\Rightarrow 3x'^2 - 4y'^2 + 0 - 0 + 3 - 4 - 6 + 8 - 10 = 0$ $\Rightarrow 3x'^2 - 4y'^2 - 9 = 0$ $\Rightarrow 3x'^2 - 4y'^2 = 9$ Now removing suffixes, the equation referred to new axes is $3x^2 - 4y^2 = 9$
12.	**Remove the first-degree terms in $3x^2 + 4y^2 - 12x + 4y + 13 = 0$**
	Solution: Given equation is $3x^2 + 4y^2 - 12x + 4y + 13 = 0$…….$(i)$ To remove x and y terms shift the origin at (α, β) without change of direction of the axes. Put $x = \alpha + x'$ and $y = \beta + y'$ Putting the values of x and y in (i), we get $3(\alpha + x')^2 + 4(\beta + y')^2 - 12(\alpha + x') + 4(\beta + y') + 13 = 0$ $\Rightarrow 3(x'^2 + 2\alpha x' + \alpha^2) + 4(y'^2 + 2\beta y' + \beta^2) - 12(\alpha + x') + 4(\beta + y') + 13 = 0$ $\Rightarrow 3x'^2 + 6\alpha x' + 3\alpha^2 + 4y'^2 + 8\beta y' + 4\beta^2 - 12\alpha - 12x' + 4\beta + 4y' + 13 = 0$ $\Rightarrow 3x'^2 + 4y'^2 + (6\alpha - 12)x' + (8\beta + 4)y' + 3\alpha^2 + 4\beta^2 - 12\alpha + 4\beta + 13 = 0 \ldots\ldots (ii)$ To remove x and y terms Put $6\alpha - 12 = 0$ and $8\beta + 4 = 0$ $\therefore \alpha = 2$ and $\beta = -\frac{1}{2}$ Putting the values of α and β in (ii), we get

	$\Rightarrow 3x'^2 + 4y'^2 + (6.2 - 12)x' + \left(8.-\frac{1}{2}+4\right)y' + 3(2)^2 + 4\left(-\frac{1}{2}\right)^2 - 12.2 + 4\left(-\frac{1}{2}\right) + 13 = 0$ $\Rightarrow 3x'^2 + 4y'^2 + 0 + 0 + 12 + 1 - 24 - 2 + 13 = 0$ $\Rightarrow 3x'^2 + 4y'^2 = 0$ Now removing suffixes, the equation referred to new axes is $3x^2 + 4y^2 = 0$
13.	**Transform the equation $17x^2 + 18xy - 7y^2 - 16x - 32y - 18 = 0$ to one in which there is no term involving x, y, xy both sets of axes being rectangular.**
	Solution: Given equation is $17x^2 + 18xy - 7y^2 - 16x - 32y - 18 = 0$ ……(i) To remove x and y terms shift the origin at (α, β) without change of direction of the axes. Put $x = \alpha + x'$ and $y = \beta + y'$ Putting the values of x and y in (i), we get $17(\alpha + x')^2 + 18(\alpha + x')(\beta + y') - 7(\beta + y')^2 - 16(\alpha + x') - 32(\beta + y') - 18 = 0$ $\Rightarrow 17(x'^2 + 2\alpha x' + \alpha^2) + 18(\alpha\beta + \alpha y' + \beta x' + x'y') - 7(y'^2 + 2\beta y' + \beta^2) - 16(\alpha + x') - 32(\beta + y') - 18 = 0$ $\Rightarrow 17x'^2 + 34\alpha x' + 17\alpha^2 + 18\alpha\beta + 18\alpha y' + 18\beta x' + 18x'y' - 7y'^2 - 14\beta y' - 7\beta^2 - 16\alpha - 16x' - 32\beta - 32y' - 18 = 0$ $\Rightarrow 17x'^2 - 7y'^2 + (34\alpha + 18\beta - 16)x' + (18\alpha - 14\beta - 32)y' + 17\alpha^2 - 7\beta^2 - 16\alpha - 32\beta + 18\alpha\beta + 18x'y' - 18 = 0 \ldots\ldots(ii)$ To remove x and y terms Put $34\alpha + 18\beta - 16 = 0$ and $18\alpha - 14\beta - 32 = 0$ Now solving these equations, we get $\therefore \alpha = 1$ and $\beta = -1$ Putting the values of α and β in (ii), we get $\Rightarrow 17x'^2 - 7y'^2 + (34.1 + 18(-1) - 16)x' + (18(-1) - 16.1 - 32)y' + 17(1)^2 - 7(-1)^2 - 16.1 - 32(-1) + 18.1.(-1) + 18x'y' - 18 = 0$ $\Rightarrow 17x'^2 + 18x'y' - 7y'^2 + 0 + 0 + 17 - 7 - 16 + 32 - 18 - 18 = 0$

	$\Rightarrow 17x'^2 + 18x'y' - 7y'^2 = 10$ Now removing suffixes, the equation referred to new axes is $17x^2 + 18xy - 7y^2 = 10$ Remove the xy term, we get $a'x'^2 + b'y'^2 - 10 = 0$ ... ... (iii) where $a = 17, b = -7, h = 9$ We know, $a' + b' = a + b = 17 - 7 = 10$ $\therefore a' + b' = 10$ $a'b' = ab - h^2 = -119 - 81 = -200$ $(a' - b')^2 = (a' + b')^2 - 4a'b'$ $= 100 + 800$ $= 900$ $\therefore a' - b' = 30$ Now solving $a' + b' = 10$ and $a' - b' = 30$ we get $a' = 20, b' = -10$ Putting the values a' and b' in (iii), we get $20\,x'^2 - 10\,y'^2 - 10 = 0$ Now removing suffixes, the equation referred to new axes is $20\,x^2 - 10\,y^2 - 10 = 0$ $\therefore 2\,x^2 - y^2 - 1 = 0$
14.	**The equation $3x^2 + 2xy + 3y^2 - 18x - 22y + 50 = 0$ is transformed to $4x^2 + 2y^2 = 1$ when referred to rectangular axes through the point $(2, 3)$. Find the inclination of the latter axes to the former.**

Solution:

Given equation is $3x^2 + 2xy + 3y^2 - 18x - 22y + 50 = 0$ ……(i)

When the origin is transferred to the point $(2, 3)$

Put $x = 2 + x'$ and $y = 3 + y'$

Putting the values of x and y in (i), we get

$3(2 + x')^2 + 2(2 + x')(3 + y') + 3(3 + y')^2 - 18(2 + x') - 22(3 + y') + 50 = 0$

$\Rightarrow 3(x'^2 + 4x' + 4) + 2(6 + 2y' + 3x' + x'y') + 3(y'^2 + 6y' + 9) - 18(2 + x') - 22(3 + y') + 50 = 0$

$\Rightarrow 3x'^2 + 12x' + 12 + 12 + 4y' + 6x' + 2x'y' + 3y'^2 + 18y' + 27 - 36 - 18\acute{x}' - 66 - 22y' + 50 = 0$

$\Rightarrow 3x'^2 + 3y'^2 + x'(12 + 6 - 18) + y'(4 + 18 - 22) + 2x'y' + 12 + 12 + 27 - 36 - 66 + 50 = 0$

$\Rightarrow 3x'^2 + 3y'^2 + 2x'y' - 1 = 0$

Now removing suffixes, the equation referred to new axes is

$\Rightarrow 3x^2 + 3y^2 + 2xy - 1 = 0$ … … … … … … … … … … … … … … … (ii)

Let the axes be rotated through an angle θ.

$$tan2\theta = \frac{2h}{a - b} = \frac{2}{3 - 3} = \infty$$

$$\therefore \theta = \frac{\pi}{4}$$

If we rotate the axes through an angle $\theta = \frac{\pi}{4}$, then we get

$$x = x'cos\frac{\pi}{4} - y'sin\frac{\pi}{4} = \frac{x' - y'}{\sqrt{2}}$$

$$y = x'sin\frac{\pi}{4} + y'cos\frac{\pi}{4} = \frac{x' + y'}{\sqrt{2}}$$

Putting the values of x and y in (ii), we get

$$3\left(\frac{x' - y'}{\sqrt{2}}\right)^2 + 3\left(\frac{x' + y'}{\sqrt{2}}\right)^2 + 2\left(\frac{x' - y'}{\sqrt{2}}\right)\left(\frac{x' + y'}{\sqrt{2}}\right) - 1 = 0$$

	$\Rightarrow 3.\frac{(x'-y')^2}{2}+3.\frac{(x'+y')^2}{2}+(x'-y')(x'+y')-1=0$ $\Rightarrow 3(x'-y')^2+3(x'+y')^2+2(x'-y')(x'+y')-2=0$ $\Rightarrow 3x'^2-6x'y'+3y'^2+3x'^2+6x'y'+3y'^2+2x'^2-2y'^2-2=0$ $\Rightarrow 8x'^2+4y'^2-2=0$ $\Rightarrow 4x'^2+2y'^2-1=0$ Now removing suffixes, the equation referred to new axes is $\Rightarrow 4x^2+2y^2=1$
15.	**Remove the first-degree terms in $3x^2-4y^2+6x+24y-135=0$**
16.	**Transform to axes inclined at 45^0 to the original axes the equation $x^2-y^2=a^2$.**
17.	**Transform the equation $x^2+6xy+9y^2+4x+8y+15=0$ to rectangular axes through the point $(-2,-1)$ and inclined at an angle $tan^{-1}\left(\frac{1}{2}\right)$.**
18.	**Transform the equation $3x^2-24xy+10y^2+6x+52y=0$ to rectangular axes through the point $(3,1)$ and inclined at an angle $tan^{-1}\left(\frac{3}{4}\right)$.**
19	**Transform the equation $14x^2-4xy+11y^2-44x-58y+71=0$ to rectangular axes through the point $(2,3)$ and inclined at an angle $tan^{-1}(2)$.**
20.	**Show that if the origin is transferred to $(0,1)$ and the axes are rotated through 45^0, the equation $5x^2-2xy+5y^2+2x-10y-7=0$ referred to the new axes becomes $4x^2+6y^2=12$.**
21.	**Show that the equation $x^2-2xy+y^2+x-3y=0$ by transferring to axes through the point $(-1,0)$ inclined at an angle of 45^0 with the original axes becomes $\sqrt{2}$ $y^2-x=0$.**
22.	**Determine the angle through which the axes must be rotated to remove the xy term in the equation $17x^2+18xy-7y^2=1$.**
23.	**Determine the angle through which the axes must be rotated to remove the xy term in the equation $3x^2+2xy+3y^2-\sqrt{2}x=0$.**
24.	**Through what angle must the axes be rotated to remove the term containing xy in $11x^2+4xy+14y^2=5$?**
25.	**By means of rotation of axes, transform the equation $x^2+4xy+4y^2-3x=6$ into an equation which has no mixed term xy.**
26.	**Transform the equation $36x^2+24xy+29y^2-72x+126y+81=0$ to one in which there is no term involving x, y, xy both sets of axes being rectangular.**